

The Quantum Sharp Young Inequality at the Plancherel Endpoint

Automated mathematics run `fa_banach.001`, lane 10

11 August 2026

Status. Candidate partial result, likely valid, pending human review. It proves one full exponent slice of Conjecture V.16, not the finite-exponent conjecture.

1 Source conjecture and result

De Palma, Trevisan, Giovannetti, and Ambrosio formulate a quantum sharp Young inequality for the n -mode beam-splitter and two-mode squeezer [1]. Their Conjecture V.16 asserts that the bilinear Schatten constant

$$C_n(p, q, r, \lambda) = \sup_{X, Y \neq 0} \frac{\|\mathcal{B}_\lambda^{\otimes n}(X \otimes Y)\|_r}{\|X\|_p \|Y\|_q}$$

is obtained by restricting to thermal Gaussian inputs and is multiplicative in n . The exact source statement and its finiteness prediction appear on PDF page 28:

Conjecture V.16 (Quantum sharp Young’s inequality). *For any $n \in \mathbb{N}$, any $p, q, r \geq 1$ and any $\lambda \geq 0$, the supremum in (72) can be restricted to thermal Gaussian states, i.e.,*

$$\begin{aligned} C_n(p, q, r, \lambda) &= \sup_{E_A, E_B \geq 0} \frac{\|\mathcal{B}_\lambda^{\otimes n}(\hat{\omega}(E_A)^{\otimes n} \otimes \hat{\omega}(E_B)^{\otimes n})\|_r}{\|\hat{\omega}(E_A)^{\otimes n}\|_p \|\hat{\omega}(E_B)^{\otimes n}\|_q} \\ &= \left(\sup_{E_A, E_B \geq 0} \frac{\|\mathcal{B}_\lambda(\hat{\omega}(E_A) \otimes \hat{\omega}(E_B))\|_r}{\|\hat{\omega}(E_A)\|_p \|\hat{\omega}(E_B)\|_q} \right)^n = C_1(p, q, r, \lambda)^n. \end{aligned} \quad (73)$$

Therefore, the constants C_n are multiplicative.

Remark V.17. We conjecture that the supremum in (73) is

- finite and achieved by finite $E_A, E_B \geq 0$ if $\frac{1}{p} + \frac{1}{q} > 1 + \frac{1}{r}$;
- finite and asymptotically achieved in the limit $E_A, E_B \rightarrow \infty$ if $\frac{1}{p} + \frac{1}{q} = 1 + \frac{1}{r}$;
- infinite and asymptotically achieved in the limit $E_A, E_B \rightarrow \infty$ if $\frac{1}{p} + \frac{1}{q} < 1 + \frac{1}{r}$.

Figure 1: Conjecture V.16 and Remark V.17 in [1].

We prove the conjecture on the Plancherel endpoint $(p, q, r) = (2, 2, \infty)$, including both kinds of Gaussian mixing and every mode number.

2 Proof intuition

A Gaussian mixing unitary acts linearly on phase-space displacements. The Weyl characteristic function of the output is therefore the pointwise product of two rescaled input characteristic functions. At Schatten exponent two, the Weyl transform is exactly Plancherel. Cauchy–Schwarz bounds the L^1 norm of that product, and Weyl inversion bounds the output operator norm. The rescaling Jacobians produce the exact constant. High-energy thermal states, with their two energies chosen in the inverse ratio of the mixing weights, meet the constant asymptotically.

3 The exact endpoint

Theorem 1 (Plancherel endpoint of Conjecture V.16). *Let $n \geq 1$.*

(i) *If $0 < \lambda < 1$ (beam-splitter), then*

$$C_n(2, 2, \infty, \lambda) = [\lambda(1 - \lambda)]^{-n/2}.$$

(ii) *If $\lambda > 1$ (two-mode squeezer), then*

$$C_n(2, 2, \infty, \lambda) = [\lambda(\lambda - 1)]^{-n/2}.$$

(iii) *At $\lambda = 0$ or $\lambda = 1$, the constant is infinite.*

For every nondegenerate λ , the sharp value is approached by thermal Gaussian inputs. In particular,

$$C_n(2, 2, \infty, \lambda) = C_1(2, 2, \infty, \lambda)^n,$$

so Conjecture V.16 and the critical-case prediction of Remark V.17 hold on this exponent slice.

4 Proof

For $z \in \mathbb{C}^n$, let $D(z)$ be the n -mode Weyl displacement and set

$$\chi_X(z) = \text{Tr}[XD(z)], \quad d\nu(z) = \pi^{-n} d^{2n}z.$$

With this normalization, Weyl inversion and Plancherel are

$$X = \int_{\mathbb{C}^n} \chi_X(z) D(-z) d\nu(z), \tag{1}$$

$$\int_{\mathbb{C}^n} |\chi_X(z)|^2 d\nu(z) = \|X\|_2^2. \tag{2}$$

These standard identities are initially read on a convenient dense class and extend to Hilbert–Schmidt operators; see, for example, Folland [2].

Let $Z = \mathcal{B}_\lambda^{\otimes n}(X \otimes Y)$. The action of the Gaussian mixing unitary on displacement operators gives, when $0 < \lambda < 1$,

$$\chi_Z(z) = \chi_X(\sqrt{\lambda}z) \chi_Y(\sqrt{1-\lambda}z). \tag{3}$$

For the squeezer $\lambda > 1$, the corresponding identity is

$$\chi_Z(z) = \chi_X(\sqrt{\lambda}z)\chi_Y(-\sqrt{\lambda-1}\bar{z}). \quad (4)$$

The reflection and conjugation in the second factor preserve Lebesgue measure.

Write $a = \lambda$ and $b = |1 - \lambda|$. Equations (3)–(4), Weyl inversion, unitarity of $D(z)$, Cauchy–Schwarz, and (2) yield

$$\begin{aligned} \|Z\|_\infty &\leq \int |\chi_X(\sqrt{a}z)\chi_Y(\sqrt{b}z)| d\nu(z) \\ &\leq \left(\int |\chi_X(\sqrt{a}z)|^2 d\nu(z) \right)^{1/2} \left(\int |\chi_Y(\sqrt{b}z)|^2 d\nu(z) \right)^{1/2} \\ &= a^{-n/2} b^{-n/2} \|X\|_2 \|Y\|_2. \end{aligned}$$

For finite-rank X, Y this is a direct estimate of the physical partial-trace formula. Since finite-rank operators are dense in the Hilbert–Schmidt class, the bound also defines and uniquely extends the bilinear map on $S_2 \times S_2$ with values in bounded operators. This proves the required upper bound.

It remains to prove sharpness. The one-mode thermal state of mean photon number E satisfies

$$\|\widehat{\omega}(E)\|_2 = (2E + 1)^{-1/2}, \quad \|\widehat{\omega}(E)\|_\infty = (E + 1)^{-1}.$$

Products of thermal states remain thermal under each modewise mixing. For a beam-splitter their output energy is

$$E_{\text{out}} = \lambda E_A + (1 - \lambda) E_B,$$

whereas for a squeezer it is

$$E_{\text{out}} = \lambda E_A + (\lambda - 1)(E_B + 1).$$

Consequently the thermal quotient for n modes equals

$$\frac{[(2E_A + 1)(2E_B + 1)]^{n/2}}{(E_{\text{out}} + 1)^n}. \quad (5)$$

Set $E_A = bt$ and $E_B = at$. In both cases, as $t \rightarrow \infty$, the expression in (5) converges to $(ab)^{-n/2}$. This matches the upper bound, proves sharpness, and proves multiplicativity.

At $\lambda = 1$, the mixing is degenerate: the retained output is the first input multiplied by the trace of the second. Taking the first thermal input to be vacuum and sending the second energy to infinity makes the quotient diverge because its Hilbert–Schmidt norm tends to zero. The symmetric argument applies at $\lambda = 0$. This completes the proof.

In particular, thermal inputs still give the correct value $+\infty$ at the two degenerate parameters, so the restriction equality in Conjecture V.16 is not contradicted. However, the finite-critical-case statement in Remark V.17 is literally false at $\lambda = 0, 1$ and should be read with nondegenerate mixing assumed.

5 Verification, scope, and upgrade obstruction

The script `code/check_endpoint.py` constructs the exact beam-splitter isometry on a four-level input Fock sector without truncating its possible output photon number. It checks 300 deterministic random matrix pairs at each of three transmissivities and checks three beam-splitter and three squeezer thermal asymptotics. It reports

PASS: Plancherel-endpoint sanity checks completed

This is supplementary verification, not the proof.

The result does not settle Conjecture V.16 at finite critical exponents, the $p \rightarrow q$ Gaussian-maximizer conjecture, the Entropy Photon-number Inequality, or multimode constrained minimum output entropy. The same characteristic function route at finite exponents requires sharp Hausdorff–Young estimates for the Weyl transform. Ordinary interpolation proves nonsharp boundedness but produces a constant strictly larger than the thermal value, so it cannot upgrade the endpoint proof to the full conjecture.

6 Novelty check and review recommendation

Bounded searches on 11 August 2026 covered the run indexes and open web/arXiv results using arXiv:1803.02360, the exact conjecture name, the authors, the phrases “ $C_n(2, 2, \infty)$,” “Hilbert–Schmidt to operator norm,” “Weyl Plancherel,” and close beam-splitter/squeezer variants. They found known one-mode, $p = 1$, $p = q$, Gaussian-input, and related partial results, but no explicit statement of this endpoint. Novelty confidence is moderate rather than high because the proof is short and may be folklore.

Human review is recommended, focusing on phase-space normalization, the finite-rank density extension, and the squeezer characteristic-function identity.

References

- [1] G. De Palma, D. Trevisan, V. Giovannetti, and L. Ambrosio, *Gaussian optimizers for entropic inequalities in quantum information*, J. Math. Phys. 59 (2018), 081101; arXiv:1803.02360.
- [2] G. B. Folland, *Harmonic Analysis in Phase Space*, Princeton University Press, 1989.